

Under what condition would the collecting duct be relatively impermeable to water?

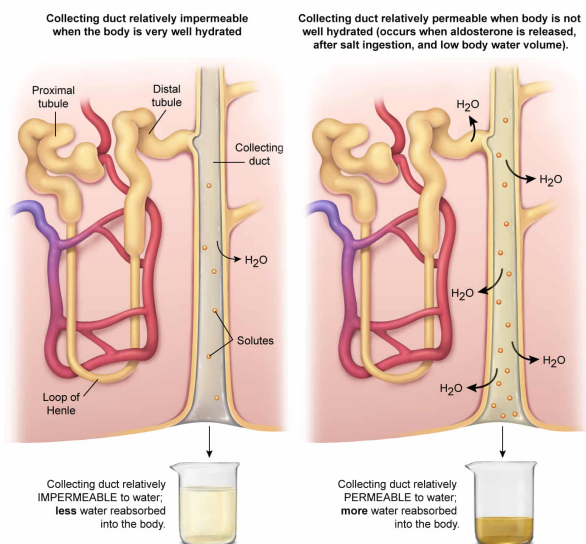
- ☐ A. When aldosterone has been released [8%]
- ☐ B. When a person has ingested large amounts of salt [7%]
- ☐ C. When water volume in the body is low [16%]
- ✓ ☒ D. When the body is very well hydrated [67%]

Correct

66% answered correctly

Explanation:

Collecting duct is relatively impermeable to water during times of hydration



Each **kidney** is divided into functional units known as **nephrons**, which produce liquid waste (urine). Blood is first **filtered** to produce a fluid known as filtrate. Water and solutes are either **secreted** into or reabsorbed out of this filtrate as the fluid moves through the **remaining portions** of the nephron.

The last portion of the nephron is known as the **collecting duct**, which mainly functions to reabsorb water back into the circulation. This reabsorption of water is primarily controlled by two hormones, **ADH** and **aldosterone**.

ADH stimulates the insertion of **aquaporins** into collecting duct cell **membranes**. This insertion increases collecting duct permeability to water, and by increasing water **reabsorption** decreases the amount of water in the urine (ie, the urine is more concentrated).

Aldosterone is released in response to decreased **blood pressure** (or blood volume), or when the serum level of K^+ is increased. Aldosterone **stimulates** the collecting duct to reabsorb Na^+ and secrete K^+ into the urine. This reabsorption of Na^+ increases the **osmolarity** of the renal interstitial fluid, which promotes water reabsorption.

This question asks for a condition in which the collecting duct would be relatively *impermeable* to water. When the volume of water in the body is abundant, it is not necessary for the collecting duct to reabsorb much water from the filtrate. Therefore, when **the body is very well hydrated**, the collecting duct would be **relatively impermeable to water**.

The other choices would cause the collecting duct to be more permeable (*not* relatively impermeable) to water.

(Choice A) Aldosterone release causes the body to reabsorb water from the collecting duct (and the distal convoluted tubule).

(Choice B) The ingestion of large amounts of salt would increase the osmolarity of the renal interstitial fluid, which *promotes* water reabsorption via **osmosis**.

(Choice C) When the water volume in the body is low (ie, a state of dehydration), ADH is released and *more* water is reabsorbed from the filtrate.

Educational objective:

The collecting ducts of the kidneys regulate water reabsorption and urine concentration. When the body is well hydrated, the collecting ducts are less permeable to water and urine is less concentrated. Conversely, during a state of dehydration, the collecting ducts are more permeable to water and urine is more concentrated.

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Subject:
Biology

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